

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-III Semester (2025-2026)
Faculty Name	Mohd. Razia Alangir Banu
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Roll Number	2520030485
Branch	CSE
Course Outcome	Apply BFS and DFS graph traversal algorithms and construct Minimum Spanning Trees using Kruskal's and Prim's algorithms to solve infrastructure design, network connectivity, and cluster analysis problems in weighted undirected graphs.

Case Study Topic: EduGraph – Adaptive Learning & Course Recommendation Platform

Work Done Summary:

Work Done Summary for CO3

Implemented BFS and DFS algorithms for course traversal and learning path analysis in the EduGraph platform. Applied Prim's and Kruskal's algorithms to create optimized course recommendation networks and improve connectivity analysis efficiently.

Implementation Details:

1. Implemented BFS algorithm for level-wise traversal of course dependency graphs.
2. Used DFS algorithm to explore learning paths and connected course structures.
3. Represented courses and prerequisites using weighted graph structures.
4. Applied Prim's algorithm to generate minimum-cost course connectivity networks.
5. Implemented Kruskal's algorithm to construct optimized recommendation networks using Minimum Spanning Trees (MST).
6. Achieved efficient graph traversal and optimized connectivity analysis with reduced processing complexity.

Result: Screenshots / Output

```
● PS C:\Users\vamshi chowdary\OneDrive\DSA\C03> javac EduGraphC03.java
● PS C:\Users\vamshi chowdary\OneDrive\DSA\C03> java EduGraphC03
BFS Traversal: 0 1 2 3 4
DFS Traversal: 0 1 3 2 4

Kruskal MST:
2 - 3 : 4
0 - 3 : 5
0 - 1 : 10

Prim MST:
0 - 1 : 10
3 - 2 : 4
0 - 3 : 5
○ PS C:\Users\vamshi chowdary\OneDrive\DSA\C03> █
```

The implementation of BFS and DFS successfully traversed the course dependency graph and explored learning paths efficiently. Kruskal's and Prim's algorithms were used to construct Minimum Spanning Trees, creating optimized course recommendation networks with minimum connection cost. The project demonstrated effective graph traversal, network connectivity analysis, and graph optimization techniques in the EduGraph platform.

GitHub Link: <https://github.com/geethika04-dot/DSA-2>

NOTE: The code and Screenshots and PDF have been uploaded in the above link, please check it there.

Student Signature:

Faculty Signature: